Swati Aggrawal

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PROFESSIONAL SUMMARY

MSc Computer Science graduate from Sapienza University of Rome specializing in Artificial Intelligence and Machine Learning. Author of an IEEE conference publication with research experience in Continual Learning and Vision Transformers. Hands-on experience developing machine learning, computer vision, and Generative AI applications using Python, PyTorch, FastAPI, Docker, Linux, and Git. Experienced in developing proof-of-concept AI solutions, building end-to-end AI pipelines, designing model evaluation frameworks, and creating AI-powered document intelligence applications. Seeking Data Scientist, AI Engineer, Machine Learning Engineer, or Applied AI roles.

SKILLS

- **Programming :** Python, SQL, C++, Bash
- **Machine Learning & AI :** PyTorch, TensorFlow, Keras, Scikit-Learn, Deep Learning, Machine Learning, Computer Vision, Generative AI, Vision Transformers, Continual Learning, Transfer Learning, Hugging Face Transformers.
- **Data Science & Analytics :** Pandas, NumPy, Matplotlib, Data Analysis, Model Evaluation, Statistical Analysis, Feature Engineering
- **Backend & APIs :** FastAPI, REST APIs
- **Tools:** Git, GitHub, Linux, Docker, AWS

EDUCATION

Sapienza Università di Roma **May 2026**

Master of Science in Computer Science

- **Relevant Coursework:** Computer Vision, Deep Learning and Applied AI, Natural Language Processing, Machine Learning, Distributed Systems
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.
- Selected for a competitive international master's program.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

RESEARCH EXPERIENCE

Continual Learning for Computer Vision (Master's Thesis) **Nov 2025**

- Evaluated pretrained Vision Transformer architectures under Task-Incremental learning settings.
- Designed experimental pipelines for training, evaluation, and comparison of computer vision models across sequential tasks.
- Investigated catastrophic forgetting and knowledge retention in transformer-based visual learning systems.
- Developed reproducible training and evaluation pipelines using Python and PyTorch.
- Automated experiment tracking and performance benchmarking across multiple continual learning scenarios.
- Documented experimental findings and identified research opportunities for future development of continual learning systems.
- **Technologies:** Python, PyTorch, Vision Transformers, Computer Vision

Machine Learning for Air Pollution [IEEE Publication] **May 2023**

- Published IEEE conference paper evaluating machine learning approaches for environmental prediction and monitoring.
- Analyzed Random Forest, SVM, Genetic Algorithm, LS-SVM, Decision Trees, and Artificial Neural Networks across multiple air quality datasets.
- Performed comparative evaluation of model performance and identified opportunities for future research and real-time monitoring applications.
- Presented scientific findings through structured research documentation and publication.

PROJECTS

DocDelta AI-Powered Document Intelligence & Comparison Platform

- Developed a document comparison and semantic change detection platform using Sentence Transformers and embedding-based similarity techniques.
- Built REST APIs using FastAPI for automated document analysis and change tracking workflows.
- Implemented semantic matching to identify added, removed, and modified content across document versions.
- Containerized the application using Docker and deployed it on AWS ECS Fargate.
- Configured application monitoring and logging using AWS CloudWatch.
- **Technologies:** Python, FastAPI, Sentence Transformers, Docker, AWS ECS Fargate, CloudWatch, REST APIs

Metal Defect Detection using Computer Vision

- Developed and evaluated computer vision models for automated metal surface defect detection using the NEU Metal Surface Defects dataset.
- Designed a comparative study of CNN, Support Vector Machine (SVM), and Random Forest approaches to analyze the effectiveness of deep learning versus traditional machine learning techniques for industrial inspection.
- Implemented image preprocessing, data augmentation, feature extraction, and model training pipelines using TensorFlow, Keras, NumPy, and Scikit-learn.
- Performed systematic model evaluation using accuracy metrics, confusion matrices, and class-level performance analysis across multiple defect categories.
- Investigated the applicability of computer vision techniques for automated visual inspection, defect classification, and quality-control systems in manufacturing environments.
- Compared deep learning and traditional machine learning approaches to identify trade-offs between accuracy, computational cost, and deployment feasibility.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Scikit-Learn

Lossy Text Compressor [Semantic NLP Pipeline]

- Investigated semantic text compression using transformer embeddings, quantization, and reconstruction techniques.
- Evaluated reconstruction quality using cosine similarity and ROUGE metrics.
- Analyzed trade-offs between compression efficiency and semantic preservation across multiple quantization levels.
- **Technologies:** Python, PyTorch, Vision Transformers